



AI MODEL FOR AUTOMATED CONTENT MODERATION:INTERGRATING HUGGING FACE MODEL FOR TEXT AND IMAGE PROCESSING

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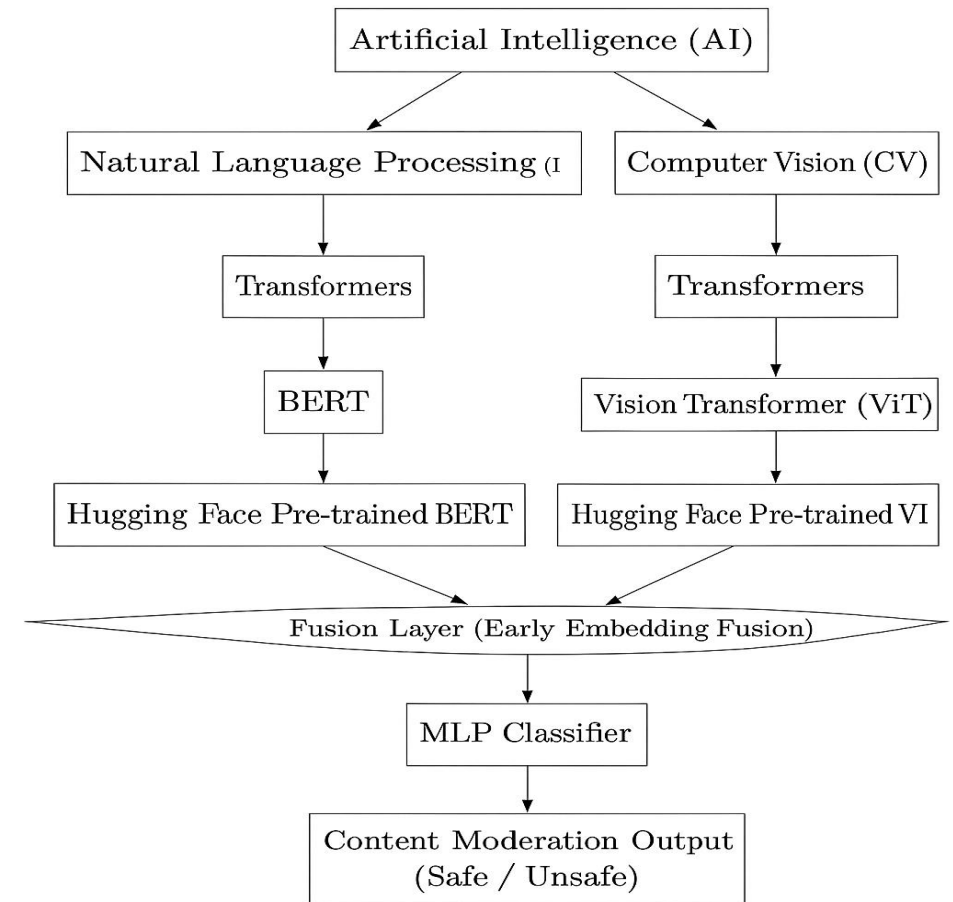
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INTRODUCTION

- This project builds an **AI-driven multimodal content moderation system** for social media.
- It integrates **BERT (text)** and **Vision Transformer (images)** with a **fusion-based classifier**.
- The model detects **toxic text, offensive memes, and unsafe imagery** in real time.
- Ensures **scalable, fair, and explainable moderation** for safe digital spaces.



PROJECT MOTIVATION

To develop a robust, scalable, and explainable multimodal content moderation framework by integrating Hugging Face pre-trained BERT and ViT models, achieving higher accuracy than unimodal baselines.

Meta's new content policies risk fueling more mass violence and genocide

NEWS TECH CREATORS
Content moderators are organizing against Big Tech



Image: The Verge

Social-media companies decide content moderation is trending down

Alexa Conne, Meghan Bobrowsky, Jeff Horowitz, The Wall Street Journal
6 min read | 08 Jan 2025, 09:19 am IST



Mark Zuckerberg, chief executive officer of Meta Platforms Inc. (Bloomberg)

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YouTube loosens content rules, says freedom of expression can outweigh harm—controversial videos may return

UN Women

FAQs: Digital abuse, trolling, stalking, and other forms of technology-facilitated violence against women

Explore our FAQs on technology-facilitated gender-based violence, including key data and insights on digital abuse, and its impact on women...

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PROJECT MOTIVATION

Problem in Focus:

- The exponential rise of user-generated content on social media brings offensive text, toxic memes, and unsafe images.
- Manual moderation is time-consuming, biased, and non-scalable.

Need for Automation:

- Platforms require real-time, scalable, and objective moderation systems.
- Existing unimodal approaches (text-only or image-only) often fail in detecting multimodal content such as memes.

PROBLEM STATEMENT

- The explosive growth of user-generated content on social media has led to the spread of offensive language, toxic memes, and unsafe imagery. Manual moderation is slow, inconsistent, and biased, while unimodal AI approaches (text-only or image-only) often fail to detect multimodal content like memes that combine text with images.
- Hence, there is a need for a multimodal, scalable, and explainable AI framework that integrates both textual and visual analysis for accurate and fair content moderation.

OBJECTIVES

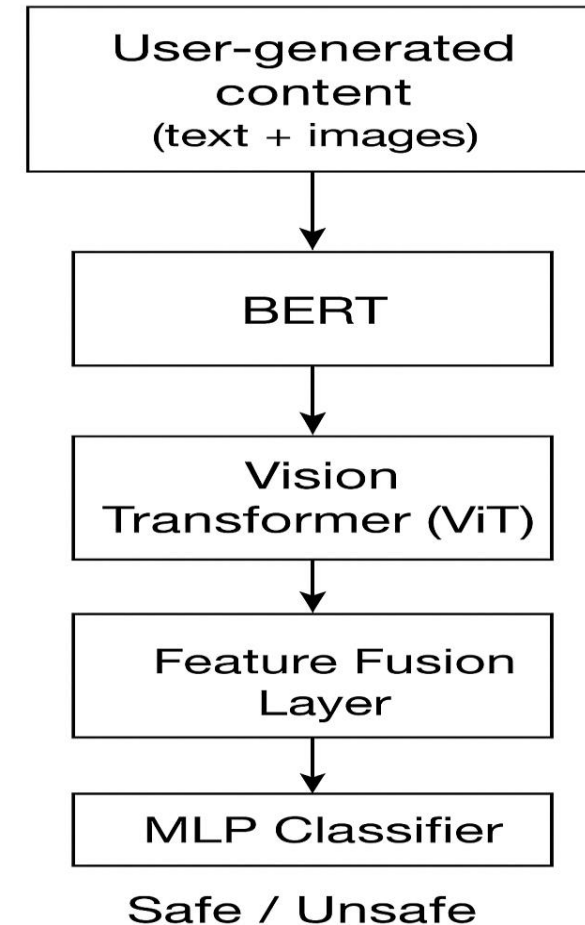
- Develop a multimodal content moderation framework:** integrating Hugging Face BERT (text) and ViT (image) models.
- Enhance detection accuracy:** by applying early fusion of embeddings for joint reasoning over text and images.
- Ensure scalability and reproducibility:** using Hugging Face's pre-trained and fine-tuned models.
- Analyze classification errors:** (false positives & false negatives) to refine the system for real-world deployment.
- Address fairness and bias issues:** by evaluating model behavior across diverse datasets and cultural contexts.

LITERATURE SURVEY

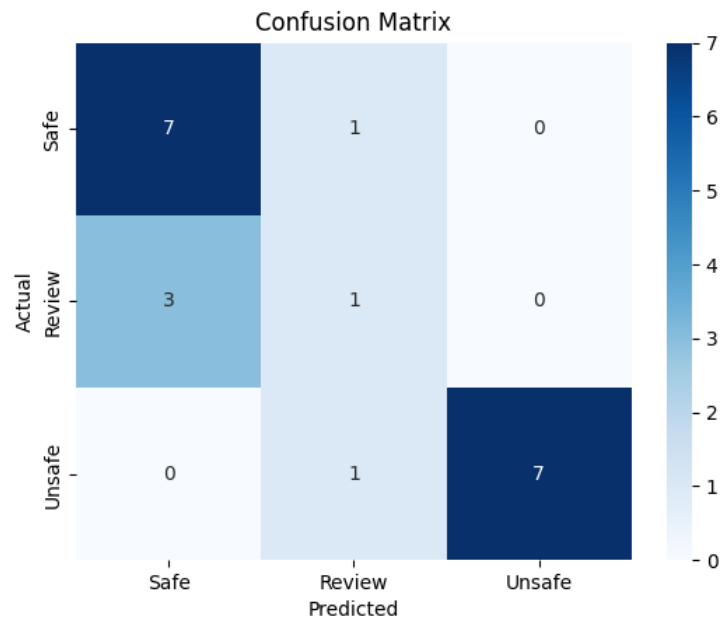
Paper Reference	Methods/Technique	Research Contributions	Research Gaps
[1] Hugging Face: Revolutionizing AI and NLP	Reviewed Hugging Face tools, libraries, and ecosystem for AI/NLP applications.	Showed how Hugging Face democratizes AI and accelerates model deployment.	Lacks experiments, performance benchmarks, and analysis of limitations or risks.
[2] Automatic Report Generation for Histopathology Images Using Pre-Trained Vision Transformers and BERT	They used HIPT for image encoding and BERT for text generation.	They proved HIPT + BERT can generate reports effectively.	Accurate report generation needs stronger vision–language alignment, since simple fusion can miss critical image–text cues.
[3] Optimizing Multimodal Transformers for Medical Image Captioning: Enhancing Automated Descriptions via AI Systems	Used a multimodal transformer with attention to generate medical image captions.	Improved caption accuracy by integrating visual and textual features.	Limited generalization across modalities and real-world clinical settings.

METHODOLOGY

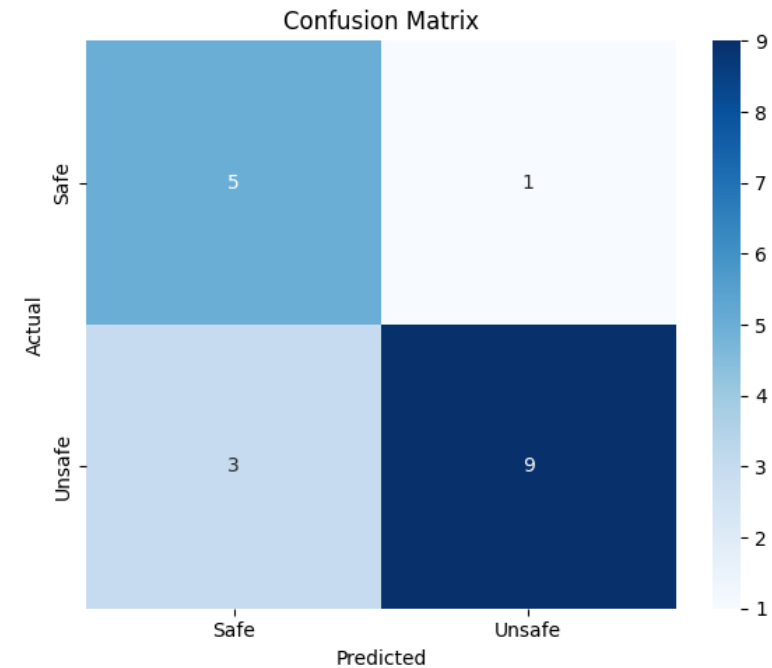
- The framework uses **two parallel pipelines**: BERT (for text analysis) and ViT (for image analysis).
- Outputs from both pipelines are combined through an **early fusion layer** and passed to an **MLP classifier**.
- Final system provides **content moderation output (Safe / Unsafe)** with improved accuracy compared to unimodal models.



RESULTS AND DISCUSSION



Text



Image

PROJECT OUTCOMES

- **Improved detection performance** compared to unimodal baselines, with higher accuracy, precision, recall, and F1-score.
- **Reduced classification errors**, especially in sarcasm and meme-based content.
- **Technical Contribution:** Demonstrated the effectiveness of Hugging Face multimodal fusion for safety-critical AI tasks.
- **Societal Impact:** Supports safer and inclusive online communities by minimizing harmful, offensive, and policy-violating content.
- **Foundation for Future Research:** Opens pathways for bias mitigation, multilingual support, and cross-modal attention techniques.

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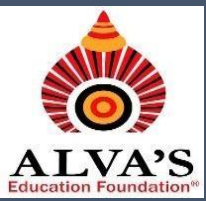
CONCLUSION AND FUTURE SCOPE

Conclusion:

- Built a multimodal moderation system (BERT + ViT) with Hugging Face.
- Outperformed unimodal baselines in accuracy and F1-score.
- Reduced false positives/negatives, improving reliability.

Future Scope:

- Add cross-modal attention for better text–image reasoning.
- Expand to multilingual and diverse datasets.
- Apply fairness-aware training to reduce bias.
- Enable real-time deployment on social media platforms.



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THANK YOU!!